

Vanilla policy gradient approach

- fundamental algorithm for policy based methods
- Rarely useful in a realistic problem settings
- important to build intuition & theoretical background for more complex algorithms.

→ Value based Methods → good action values } Policy is obtained

Policy gradient Method → optimizing policy wrt RL objective.

* $J(\theta) \leftarrow \sum P(z; \theta) \left[\sum_{t=0}^T \gamma^t R(s_t, A_t) \right] R(z)$

$J(\pi_\theta) = E_{z \sim p_\theta(z)} \left[\sum_{t=0}^T \gamma^t R(s_t, A_t) \right]$

$J \rightarrow$ objective and it is a function of policy (π_θ) at hand.

[* Here when we parametrized a policy we will explicitly model π . Learning is now done by approximate gradient ascent updates.]

$z \rightarrow$ trajectory the agent observes, it is random with the probability distribution p_θ . It is a fn of Policy hence fn of θ .

$R \rightarrow$ fn (unknown to agent) that gives a reward based on the environment dynamics given state s_t & action A_t .

$R(z) \rightarrow$ cumulative Return from trajectory.

* we have to maximize objective fn J which depends on θ .

→ A Natural thing to do is take a gradient step in the ascending direction: $\rightarrow$ step size

$$\theta_{k+1} = \theta_k + \alpha \nabla_\theta J(\pi_{\theta_k})$$

$\nabla_\theta J(\pi_{\theta_k}) \rightarrow$ Policy gradient term [we need to find this term]

$\nabla_\theta \rightarrow$ gradient w.r.t θ .

Why Equation of objective fn is :-

$$J(\theta) = \sum_{\tau} P(\tau; \theta) R(\tau)$$

①
Cumulative return from trajectory.

Probability of the trajectory
(depends on θ since it defines the policy that it uses to select the actions of trajectory which as an impact of the states visited)

We calculate Expected return $J(\theta)$ by summing for all trajectories the probability of taking that trajectory given θ & the return of this trajectory.

~~from~~
$$P(\tau; \theta) = \prod_{t=0} P(s_{t+1} | s_t, a_t) \pi_{\theta}(a_t | s_t)$$

Environment dynamics (state distribution)

②

Probability of taking that action a_t at state s_t .

→ Chain of product of ~~probabilities~~ probabilities of observing a state, taking a particular action a_t the policy given the state & observing the next state.

from Equ ① & ② we can write :-

~~$J(\theta) = \sum_{\tau} P(\tau; \theta) R(\tau)$~~

In Equ ① we can rewrite as

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \sum_{\tau} P(\tau; \theta) R(\tau)$$

we can rewrite the gradient ~~from~~ as (3)

$$\nabla_{\theta} J(\theta) = \sum_{\tau} \nabla_{\theta} P(\tau; \theta) R(\tau)$$

[we then multiply every term in the sum by $\frac{P(\tau; \theta)}{P(\tau; \theta)}$]

$$= \sum_{\tau} \frac{P(\tau; \theta)}{P(\tau; \theta)} \nabla_{\theta} P(\tau; \theta) R(\tau)$$

$$= \sum_{\tau} \frac{\nabla_{\theta} P(\tau; \theta)}{P(\tau; \theta)} R(\tau)$$

$$\nabla_{\theta} J(\theta) = \sum_{\tau} P(\tau; \theta) \nabla_{\theta} \log P(\tau, \theta) R(\tau) \quad (2)$$

derivative log trick (also called likelihood ratio trick) a simple rule of calculus that implies that $\nabla_x \log f(x) = \frac{\nabla_x f(x)}{f(x)}$

so given, $\frac{\nabla_{\theta} P(\tau; \theta)}{P(\tau; \theta)}$ will be $\nabla_{\theta} \log P(\tau | \theta)$

Thanks to this formula (2) we can estimate the gradient from the batch of m trajectories :-

$$\nabla_{\theta} J(\theta) = \frac{1}{m} \sum_{i=1}^m \nabla_{\theta} \log P(\tau^i; \theta) R(\tau^i)$$

where τ^i is sampled ~~trajectories~~ trajectory.

By taking avg. $(\frac{1}{m} \sum_{i=1}^m)$; you obtain an estimate that is unbiased w.r.t. the true gradient } { avg. reduces the variance of gradient estimate }

Simplifying Eq. (2)

$$\nabla_{\theta} J(\theta) = \sum_{\tau} P(\tau; \theta) \nabla_{\theta} \log P(\tau, \theta) R(\tau)$$

simp by this expression further.

$$\nabla_{\theta} \log P(\tau, \theta) = \nabla_{\theta} \log \left[\mu_0(s_0) \prod_{t=0}^{\infty} p(s_{t+1} | s_t, a_t) \pi_{\theta}(a_t | s_t) \right] \quad (4)$$

where $\mu_0(s_0)$ $\rightarrow$ initial distribution & $p(s_{t+1} | s_t, a_t)$ is state transition dynamics.

Since the initial distribution & state transition dynamics are independent of θ ; they can be dropped:-

$$\text{So, } \nabla \log P(\tau, \theta) = \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

So, we can rewrite our (2) Eq. as:-

$$\boxed{\nabla_{\theta} J(\theta) = P(\tau; \theta) \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(\tau)}$$

$\uparrow$
Policy gradient Expression.

Note

* Initial state distribution:-

$\rightarrow$ Since the policy parameter θ do not affect the initial state when the task is starting hence derivative is zero.

* State transition dynamics: This represents the policy parameter θ do not affect these transition probabilities; so derivative will be zero.